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MATH 246A

Homework 3
Due: TBD

Problem 5.1

Show that the infinite series

$$\frac{1}{z} + \sum_{n \in \mathbb{N}} \frac{2z}{z^2 - n^2}$$

converges absolutely for each $z \in \mathbb{C} \setminus \mathbb{Z}$.

Solution. Let $z \in \mathbb{C} - \mathbb{Z}$ be arbitrary. Note that:

$$\lim_{n \to \infty} \left| \frac{z^2 - n^2}{n^2} \right| = 1.$$

Thus there exists a $N \in \mathbb{N}$ such that for all $n \geq N$:

$$-\frac{1}{2} < \left| \frac{z^2 - n^2}{n^2} \right| - 1 < \frac{1}{2} \Rightarrow \left| \frac{z^2 - n^2}{n^2} \right| > \frac{1}{2} \Rightarrow \left| z^2 - n^2 \right| > \frac{1}{2} n^2 \Rightarrow \frac{1}{|z^2 - n^2|} \leq \frac{2}{n^2}.$$

Thus we have the following

$$|z|^{-1} + \sum_{k=1}^{\infty} \frac{2|z|}{|z^2 - n^2|} = |z|^{-1} + \sum_{k=1}^{N-1} \frac{2|z|}{|z^2 - n^2|} + \sum_{k=N}^{\infty} \frac{2|z|}{|z^2 - n^2|}$$

Thus

$$\sum_{k=1}^{\infty} \frac{2z}{z^2 - n^2} \text{ converges absolutely} \Longleftarrow \sum_{k=N}^{\infty} \frac{2|z|}{|z^2 - n^2|} \text{ converges}$$

The latter sum converges because

$$\sum_{k=N}^{\infty} \frac{2|z|}{|z^2 - n^2|} \leq 4|z| \sum_{k=N}^{\infty} \frac{1}{n^2}$$

The second sum converges by the P test.

Problem 5.2

Let Γ be a rank-2 lattice in $\mathbb{C}$, that is, a set of the form

$$\Gamma = \{n\omega_1 + k\omega_2 : k, n \in \mathbb{Z}\},$$

where $\omega_1, \omega_2 \in \mathbb{C} \setminus \{0\}$ are linearly independent over $\mathbb{R}$. Show that

$$\sum_{\gamma \in \Gamma \setminus \{0\}} \frac{1}{\gamma^n}$$

converges absolutely for $n \in \mathbb{N}$, $n \geq 3$.

Solution.

$$S = \sum_{\gamma \in \Gamma - \{0\}} \left| \frac{1}{(\gamma)^k} \right| = \sum_{(n,m) \in \mathbb{Z}^2 - \{0\}} \frac{1}{|n\omega_1 + m\omega_2|^k}.$$

Let

$$\begin{aligned} \|\cdot\|_s := \mathbb{R}^2 &\to (0, \infty) \\ (x, y) &\to |x\omega_1 + y\omega_2|. \end{aligned}$$

One can check that $\|\cdot\|_s$ is a norm and since all norms are equivalent on $\mathbb{R}^n$, thus $\exists C \in \mathbb{R}$ where $\|\cdot\|_s \geq C\|\cdot\|_\infty$. Thus

$$\begin{aligned} S &\leq \frac{1}{C^k} \sum_{(n,m) \in \mathbb{Z}^2 - \{0\}} \frac{1}{\|(n,m)\|_\infty^k} \\ &= \frac{1}{C^k} \sum_{(m,n) \in \mathbb{Z}^2 - \{0\}} \frac{1}{\max(|n|, |m|)^k} = \frac{1}{C^k} T. \end{aligned}$$

We want to examine sum of T.

$$\begin{aligned} \sum_{n,m \in \mathbb{Z} - \{0\}} \frac{1}{\max(|n|, |m|)^k} &= 4 \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{\max(|n|, |m|)^k} \\ &= 4 \sum_{m=1}^{\infty} \left(\frac{m-1}{m^k} + \sum_{n=m}^{\infty} \frac{1}{|n|^k} \right) \\ &= 4 \sum_{m=1}^{\infty} \left(\frac{1}{m^{k-1}} + \sum_{n=m+1}^{\infty} \frac{1}{n^k} \right) \end{aligned}$$

Note that

$$\sum_{m=1}^{\infty} \frac{1}{m^{k-1}} = \zeta(k-1)$$

which is absolutely convergent if and only $k - 1 \geq 2 \Rightarrow k \geq 3$.

$$\begin{aligned} \sum_{m=1}^{\infty} \sum_{n=m+1}^{\infty} \frac{1}{n^k} &= \sum_{n \geq 2} \frac{\#\{m : 1 \leq m \leq n-1\}}{n^k} \\ &= \sum_{n \geq 2} \frac{n-1}{n^k} = \zeta(k-1) - \zeta(k) \end{aligned}$$

Thus

$$T = 8\zeta(k-1) - 4\zeta(k) + 4\zeta(k) = 8\zeta(k-1)$$

which is only convergent for $k - 1 \geq 2 \Rightarrow k \geq 3$

Problem 5.3

Show that the infinite product

$$\prod_{n=1}^{\infty}\left(1+\frac{1}{n}\right)e^{-\frac{1}{n}}$$

converges.

Solution. Note that all factors of the product are positive. Let

$$p_k := \prod_{n=1}^{k}\left(1+\frac{1}{n}\right)e^{-\frac{1}{n}}.$$

To show convergence of the limit we will show p_k is monotonic. To do this we will examine:

$$f(x) = \left(1+\frac{1}{x}\right)e^{-\frac{1}{x}}.$$

$$\begin{aligned} f'(x) &= -\frac{1}{x^2}e^{-\frac{1}{x}} + \left(1+\frac{1}{x}\right)e^{-\frac{1}{x}} * \left(\frac{1}{x^2}\right) \\ &= \frac{1}{x^2}e^{-\frac{1}{x}}\left(1+\frac{1}{x}-1\right) \\ &= \frac{1}{x^2}e^{-\frac{1}{x}}\left(\frac{1}{x}\right) \\ &= \frac{1}{x^3}e^{-\frac{1}{x}}. \end{aligned}$$

Note that $f'(x) > 0$ for all $x > 0$. Thus on $(0,\infty)$, f is monotone increasing and hence $\left(1+\frac{1}{n}\right)e^{-\frac{1}{n}}$ is monotone increasing. Furthermore,

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)e^{-\frac{1}{n}} = 1 \Rightarrow \left(1+\frac{1}{n}\right)e^{-\frac{1}{n}} \leq 1 \Rightarrow p_k \geq p_{k+1}.$$

Hence p_k is monotone decreasing and bounded below by 0 and hence convergent. However for the infinite product to exists

$$\lim_{n\to\infty} p_n \neq 0.$$

$$\begin{aligned} p_k &= \prod_{n=1}^{k}\left(\frac{n+1}{n}\right)e^{-\frac{1}{n}} \\ &= (k+1)e^{-\sum_{n=1}^{k}\frac{1}{n}} \end{aligned}$$

We want to bound p_k away from 0.

$$\begin{aligned}
\log(p_k) &= \sum_{n=1}^{k}\left(\log(n+1) - \log(n) - \frac{1}{n}\right) \\
&= \sum_{n=1}^{k}\left(\int_n^{n+1} \frac{1}{t}\,\mathrm{d}t - \frac{1}{n}\right) \\
&= -\sum_{n=1}^{k}\int_0^1 \frac{t}{n(n+t)}\,\mathrm{d}t \\
&\geq -\sum_{n=1}^{k}\frac{1}{n^2}\int_0^1 t\,\mathrm{d}t \\
&= -\frac{1}{2}\sum_{n=1}^{k}\frac{1}{n^2} \geq -\frac{1}{2}\sum_{n=1}^{\infty}\frac{1}{n^2} = -\frac{\pi^2}{12}
\end{aligned}$$

Note $p_k < 1$ and $\log(p_k) \geq -\frac{\pi^2}{12}$. Thus $e^{-\frac{\pi^2}{12}} \leq p_k \leq 1$. Thus the limit $k \to \infty$ of p_k, is nonzero and the product converges.

Problem 5.4

Let $b_n \geq 0$ for $n \in \mathbb{N}$. Show that the infinite product $\prod_{n=1}^{\infty}(1-b_n)$ converges if and only if $\sum_{n=1}^{\infty} b_n$ converges.

Solution. We will use Prop 5.13, to make some reductions.

$$\prod_{n=1}^{\infty}(1-b_n) = \prod_{n=1}^{\infty}(1+(-b_n)).$$

$$\sum_{i=1}^{\infty} b_i \text{ coverges} \overset{\text{Prop 5.13}}{\Leftrightarrow} \prod_{n=1}^{\infty}(1+b_n) \text{ converges} \overset{\text{Prop 5.13}}{\Rightarrow} \prod_{n=1}^{\infty}(1+(-b_n)) \text{ converges}$$

Thus we only need to show that

$$\prod_{n=1}^{\infty}(1+(-b_n)) \text{ converges} \Rightarrow \sum_{k=1}^{\infty} b_k \text{ converges.}$$

Note that

$$\log\left(\frac{1}{1-x}\right) \text{ continous at } x = 0.$$

. For all $\varepsilon > 0$, there exists a $\varepsilon_0 > 0$ such that $\forall x \in (-\varepsilon_0, \varepsilon_0)$:

$$\left|\log\left(\frac{1}{1-x}\right)\right| < \varepsilon.$$

By the convergence of the product: $\exists N \in \mathbb{N}$ such that $\forall m \geq n \geq N$ we have that

$$1 - \frac{\varepsilon_0}{2} < \prod_{k=n}^{m}(1 - b_n) \leq 1 + \frac{\varepsilon_0}{2}$$

Note that

$$1 - x \leq e^{-x} \text{ for all } x \in \mathbb{R}.$$

Thus

$$\begin{aligned}\prod_{k=n}^{m}(1 - b_n) &\leq \prod_{k=n}^{m} e^{-b_n} \\ &= e^{-\sum_{k=n}^{m} b_n} \Rightarrow \\ \sum_{k=n}^{m} b_n &\leq -\log\left(\prod_{k=n}^{m}(1 - b_n)\right).\end{aligned}$$

Thus:

$$0 \leq \sum_{k=n}^{m} b_n \leq -\log\Big(1 - \frac{\varepsilon_0}{2}\Big)$$

Since $-\log(1 - \frac{\varepsilon_0}{2}) = \left|\log\Big(\frac{1}{1-\frac{\varepsilon_0}{2}}\Big)\right| < \varepsilon$ by the choice of ε_0, we get $0 \leq S_{n,m} < \varepsilon$ for all $m \geq n \geq N$. Hence the partial sums of $\sum b_k$ are Cauchy, so $\sum_{k=1}^{\infty} b_k$ converges. ■

Problem 5.5

Let $n \in \mathbb{N}$ and $z_k \in \mathbb{C}$ for $k = 1, ..., n$. Show that there exists a set $J \subseteq \{1, ..., n\}$ such that

$$\left|\sum_{j \in J} z_j\right| \geq \frac{1}{\pi}\sum_{k=1}^{n}|z_k|.$$

Hint: For $t \in [0, 2\pi]$ define $J(t) \coloneqq \big\{j \in \{1, ..., n\} : \mathrm{Re}\big(z_j e^{-it}\big) \geq 0\big\}$. Find a lower bound for the average of $\sum_{j \in J(t)} z_j$.

Solution.

$$\begin{aligned}\frac{1}{2\pi}\int_0^{2\pi}\left|\sum_{k \in J(t)} z_k\right| \mathrm{d}t &= \frac{1}{2\pi}\int_0^{2\pi}\left|e^{-it}\sum_{k=1}^{n} z_k\right| \mathrm{d}t \\ &= \frac{1}{2\pi}\int_0^{1}\left|\sum_{k \in J(t)} e^{-it} z_k\right| \mathrm{d}t \\ &\geq \frac{1}{2\pi}\int_0^{1}\Re\left(\sum_{k \in J(t)} e^{-it} z_k\right) \mathrm{d}t\end{aligned}$$

Problem 5.6

Let $a_n \in \mathbb{C}$ for $n \in \mathbb{N}$.
(a) Show that the series $\sum_{n=1}^{\infty} a_n$ converges absolutely if and only if the following condition is true: there exists a constant $L \in \mathbb{C}$ such that for all $\varepsilon > 0$ there exists a finite set $F_0 \subseteq \mathbb{N}$ such that for all finite sets $F \subseteq \mathbb{N}$ with $F_0 \subseteq F$ we have

$$\left| \sum_{k \in F} a_k - L \right| < \varepsilon.$$

(b) Show that in this case $L = \sum_{n=1}^{\infty} a_n$.

Solution. $\Rightarrow$: Suppose

$$\sum_{n=0}^{\infty} a_n$$

converges absolutely to L. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that

$$\left| \sum_{k=1}^{n} a_n - L \right| < \frac{\varepsilon}{2}$$

for $n \geq N$. Additionally $\exists M \in \mathbb{N}$ such that

$$\sum_{k=n}^{m} |a_n| < \frac{\varepsilon}{2}$$

for $m \geq n > M$. Let $K = \max(M, N)$. Let $F_0 = \{1, ..., K\}$. $\forall F \subseteq \mathbb{N}$ finite where $F_0 \subseteq F$. $F \subseteq \{1, ..., n\}$ for $n \geq K$. Let $G = \{K+1, ..., n\} - F$.

$$\begin{aligned}
\left| \sum_{k \in F} a_k - L \right| &= \left| \sum_{k=1}^{n} a_k - \sum_{k \in G} a_k - L \right| \\
&\leq \left| \sum_{k=1}^{n} a_k - L \right| + \sum_{k \in G} |a_k| \\
&\leq \frac{\varepsilon}{2} + \sum_{k=K+1}^{n} |a_k| \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
\end{aligned}$$

$\Longleftarrow$: Suppose that for all $\varepsilon > 0$ there exists a F_ε finite such that all finite sets $F \supseteq F_\varepsilon$ we have

$$\left| \sum_{k \in F} a_k - L \right| < \varepsilon.$$

We will show any rearrangement of the sum is convergent and then by contrapositive of riemann rearrangement we are done.